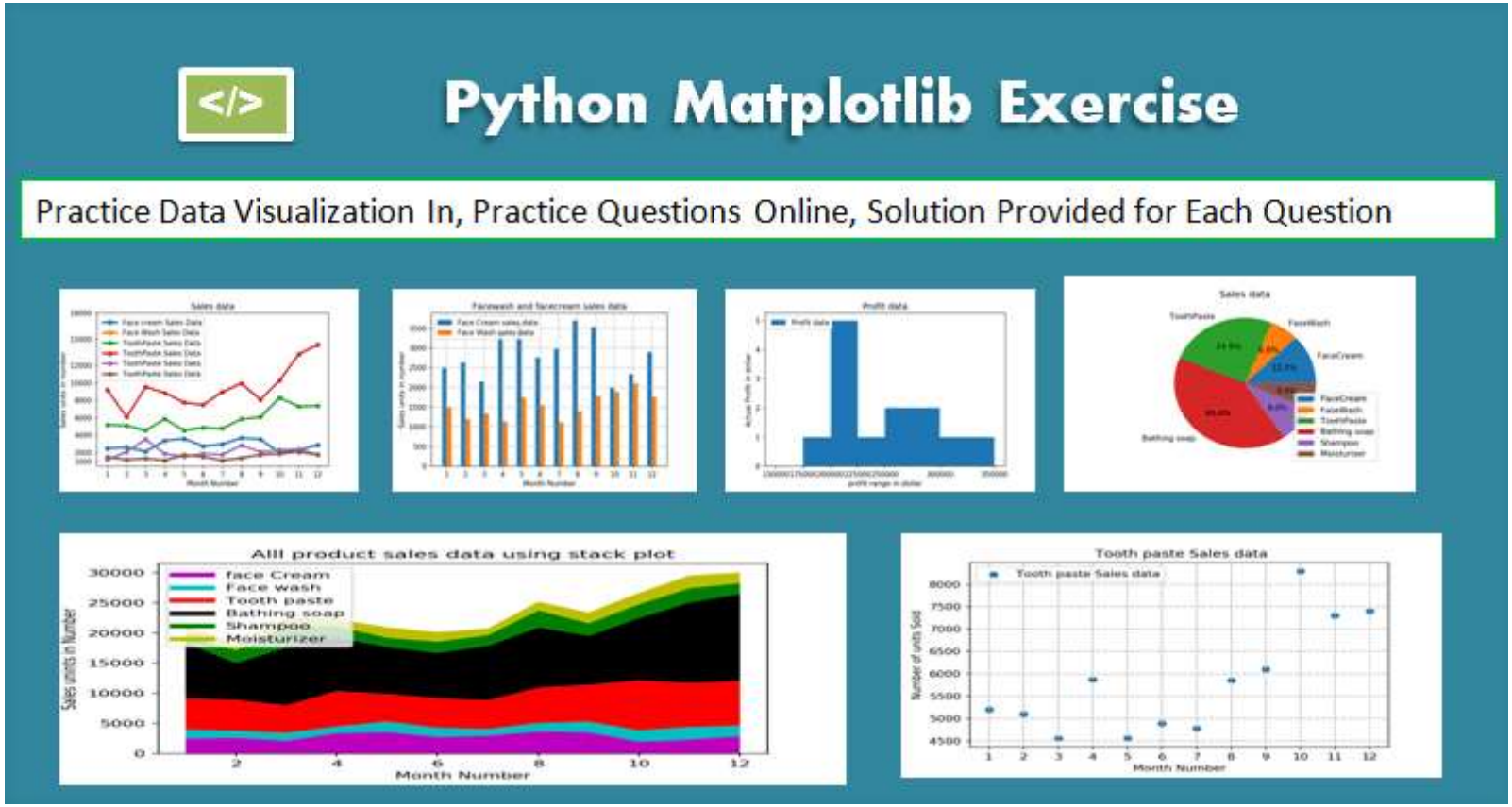
What is Matplotlib?

- Matplotlib is a comprehensive library for creating static, animated, and interactive visualizations in Python. It is an open-source and free to use.
- Plotting of data can be extensively made possible in an interactive way by Matplotlib, which is a plotting library that can be demonstrated in Python scripts.
- Plotting of graphs is a part of data visualization, and this property can be achieved by making use of Matplotlib.



In [ ]:

```
```

In [1]:

```
!pip install matplotlib
```

```
Requirement already satisfied: matplotlib in c:\users\admin\anaconda3\lib\site-packages (3.7.1)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\admin\anaconda3\lib\site-packages (from matp
lotlib) (1.0.5)
Requirement already satisfied: cycler>=0.10 in c:\users\admin\anaconda3\lib\site-packages (from matplotl
ib) (0.11.0)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\admin\anaconda3\lib\site-packages (from mat
plotlib) (4.25.0)
Requirement already satisfied: kiwisolver>=1.0.1 in c:\users\admin\anaconda3\lib\site-packages (from mat
plotlib) (1.4.4)
Requirement already satisfied: numpy>=1.20 in c:\users\admin\anaconda3\lib\site-packages (from matplotli
b) (1.24.3)
Requirement already satisfied: packaging>=20.0 in c:\users\admin\anaconda3\lib\site-packages (from matpl
otlib) (23.0)
Requirement already satisfied: pillow>=6.2.0 in c:\users\admin\anaconda3\lib\site-packages (from matplot
lib) (9.4.0)
Requirement already satisfied: pyparsing>=2.3.1 in c:\users\admin\anaconda3\lib\site-packages (from matp
lotlib) (3.0.9)
Requirement already satisfied: python-dateutil>=2.7 in c:\users\admin\anaconda3\lib\site-packages (from
matplotlib) (2.8.2)
Requirement already satisfied: six>=1.5 in c:\users\admin\anaconda3\lib\site-packages (from python-dateu
til>=2.7->matplotlib) (1.16.0)
```

In [ ]:

```
```

In [4]:

```
import numpy as np
import matplotlib.pyplot as plt
```

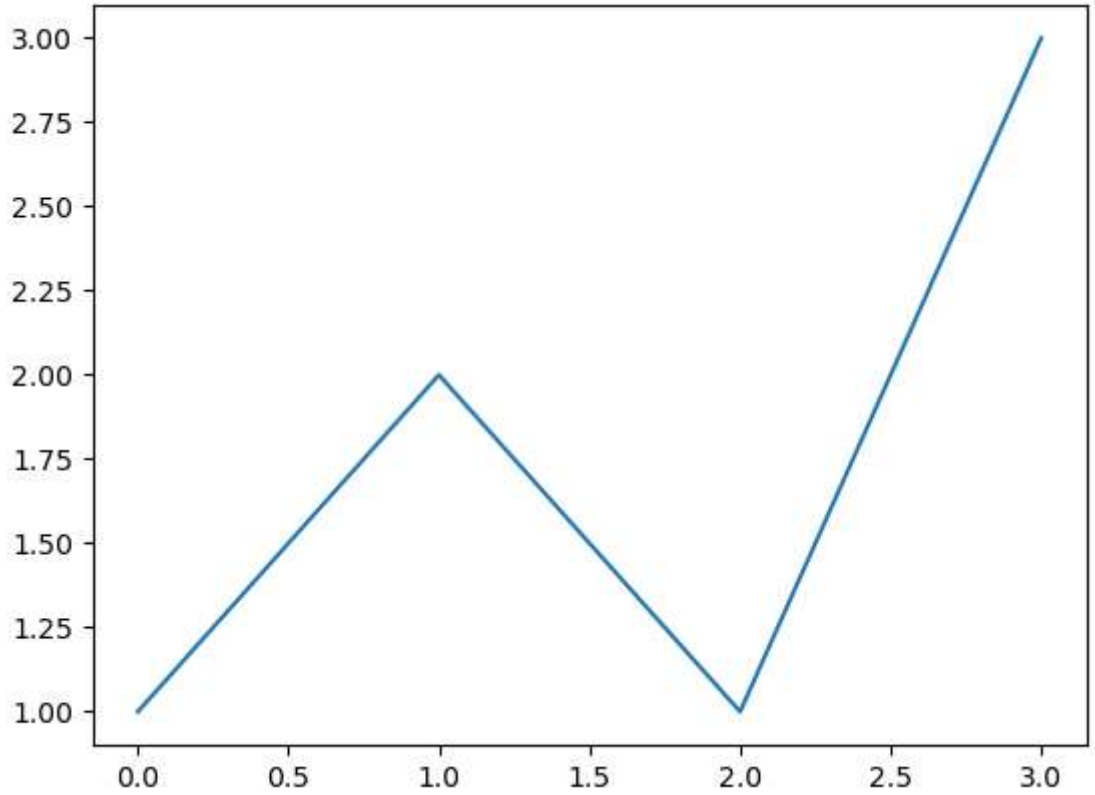
In [3]:

```
[1,2,1,3]
```

Out[3]:

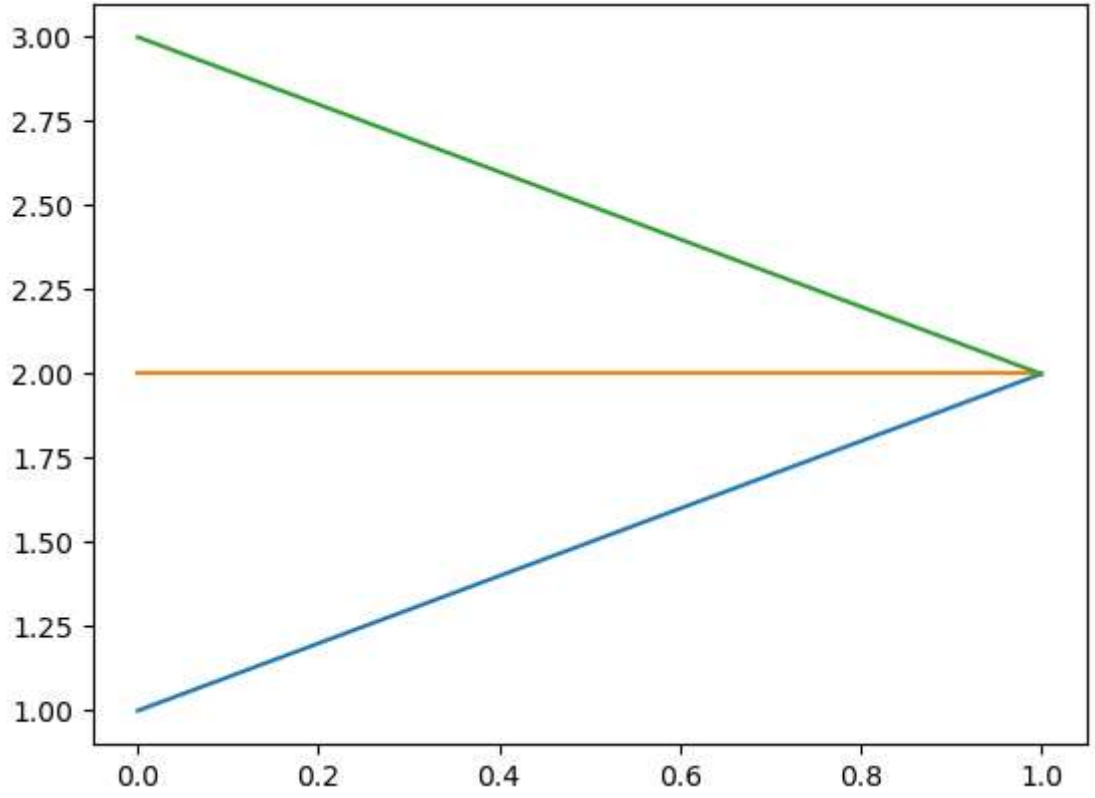
```
[1, 2, 1, 3]
```

In [5]:
```
plt.plot([1,2,1,3])
plt.show()
```

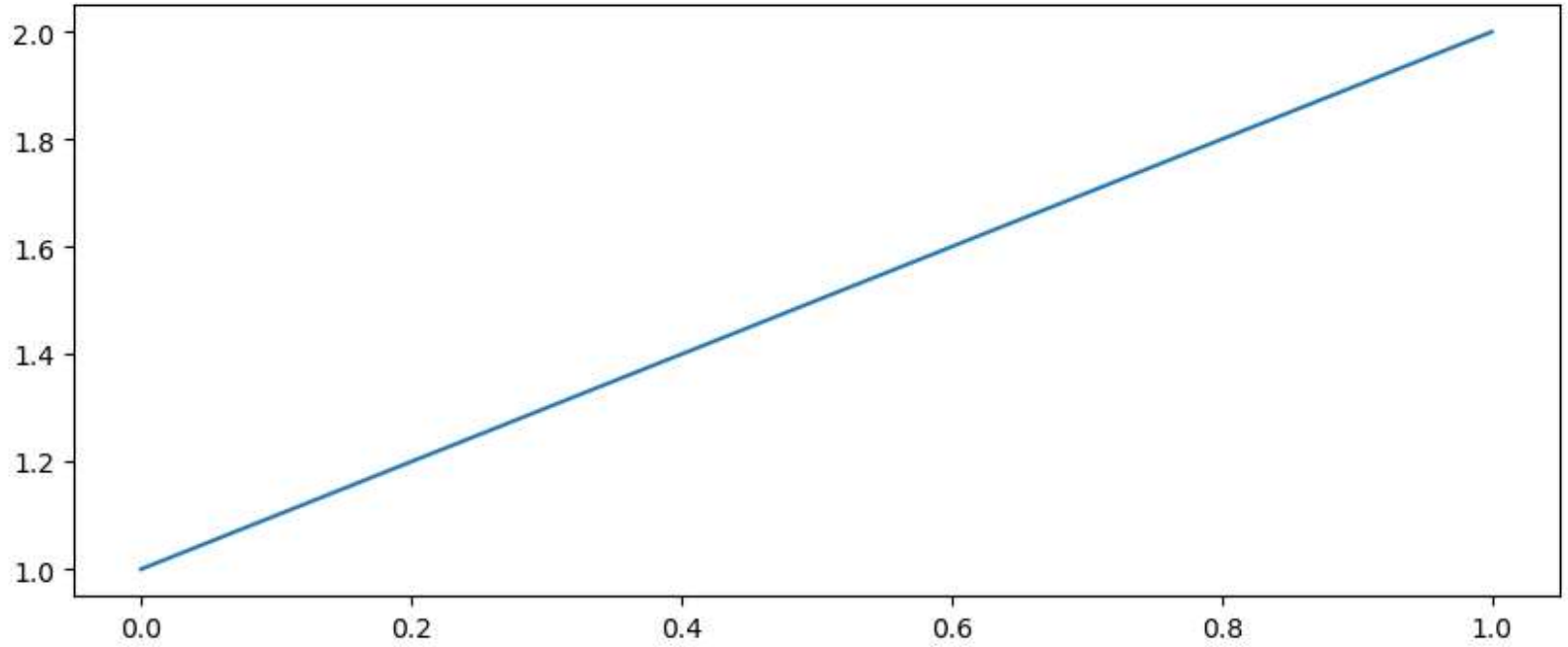


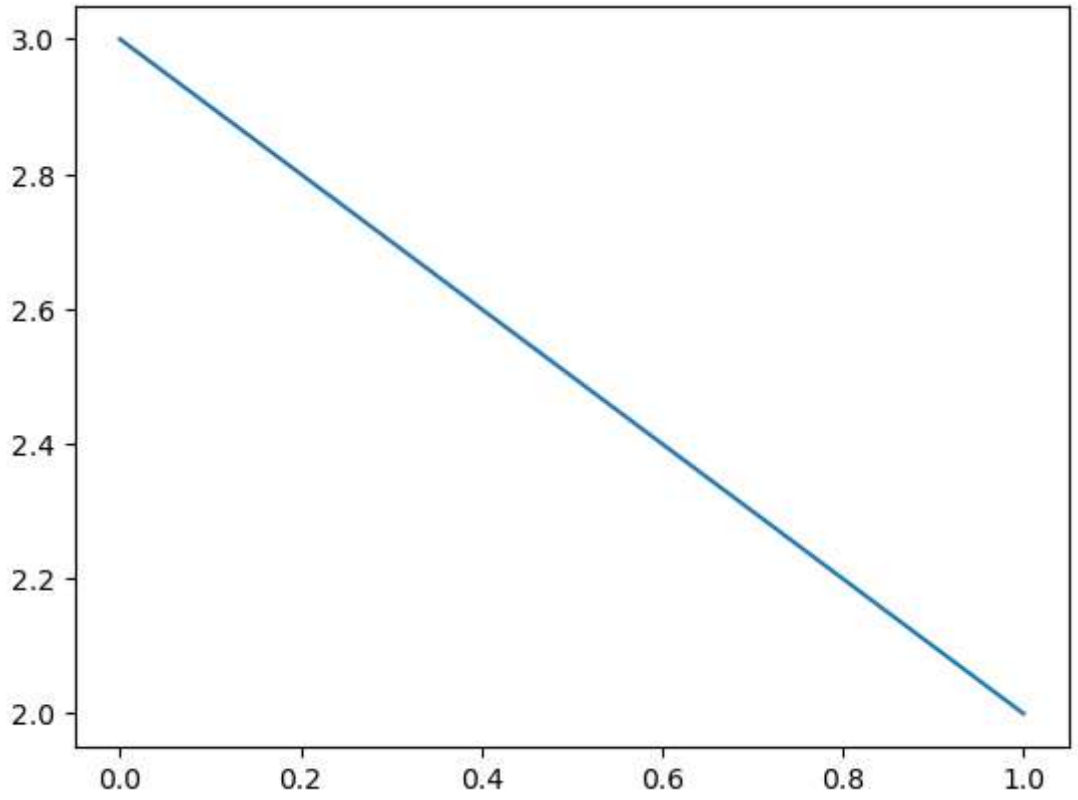
In [ ]:
```

```

In [6]:
```
plt.plot([1,2])
plt.plot([2,2])
plt.plot([3,2])

plt.show()
```



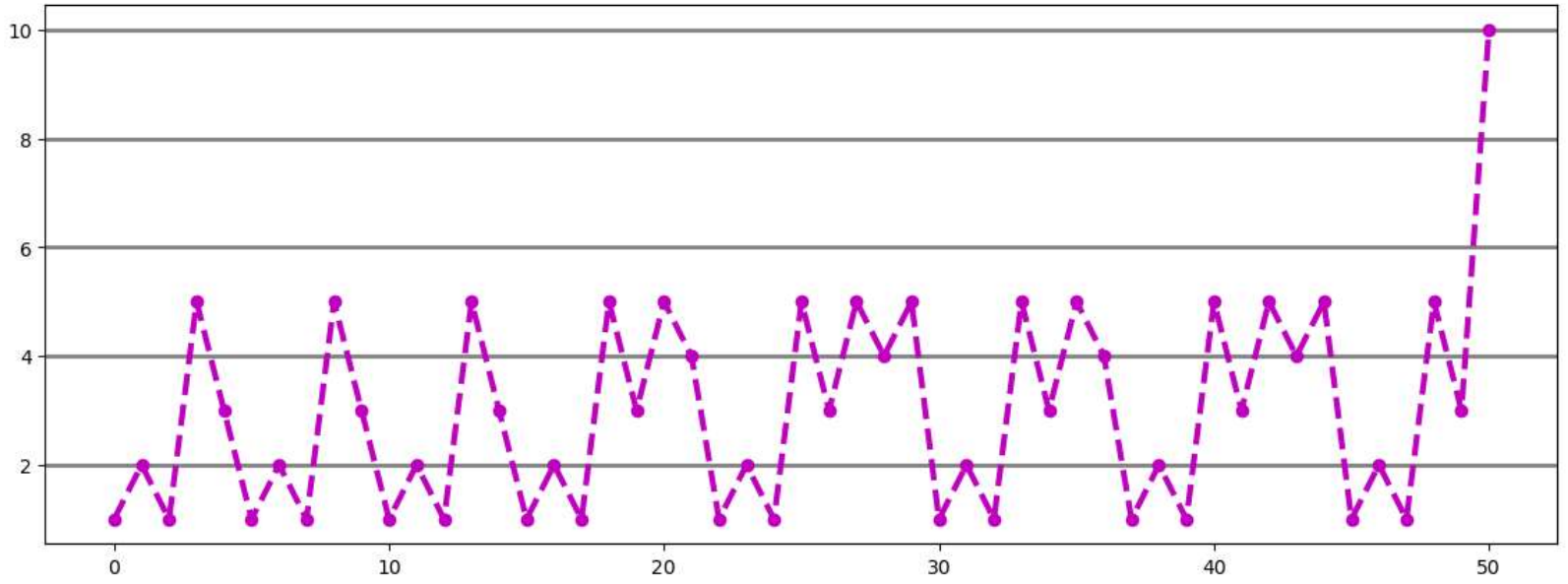
In [9]:
```
plt.figure(1, figsize=(10, 4))
plt.plot([1,2])

plt.figure(2)
plt.plot([3,2])

plt.show()
```





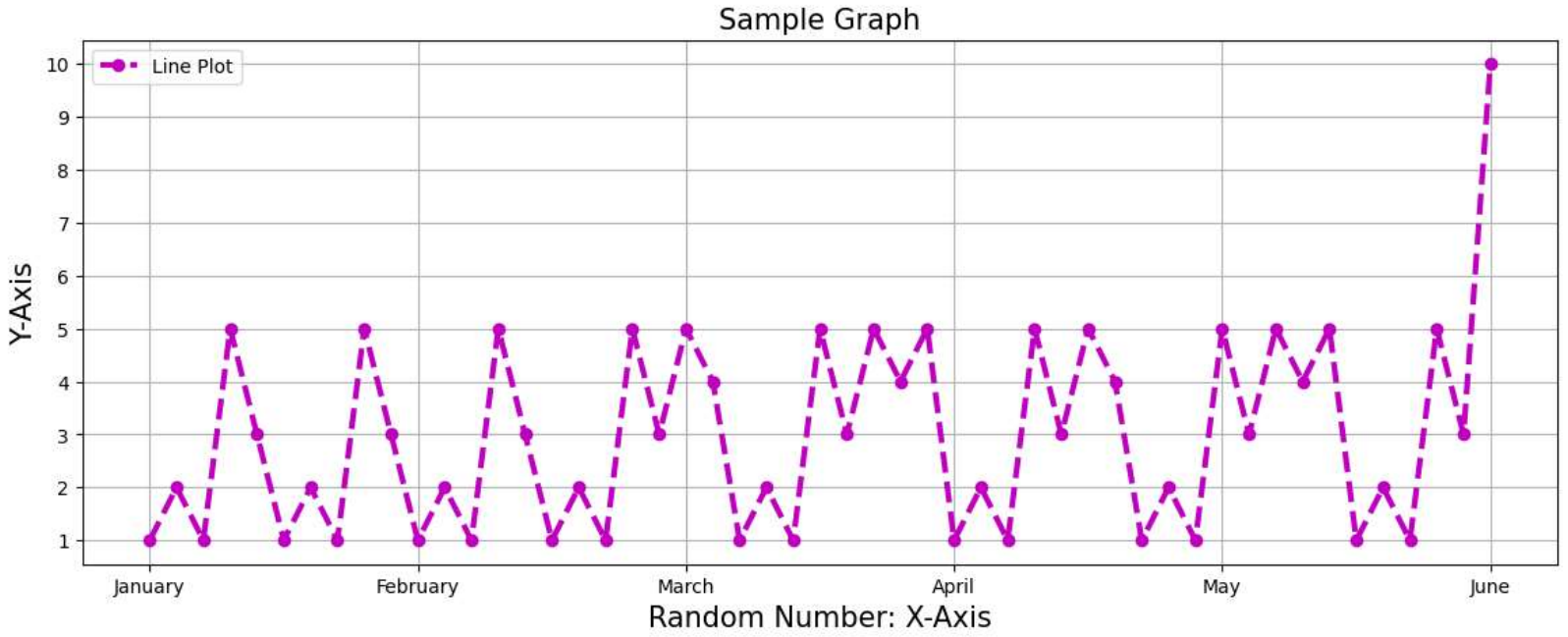
In [ ]:

In [50]:
```
plt.figure(figsize=(14, 5))
plt.plot([1,2,1,5,3,1,2,1,5,3,1,2,1,5,3,1,2,1,5,3,5,4,1,2,1,5,3,5,4,5,
          1,2,1,5,3,5,4,1,2,1,5,3,5,4,5,1,2,1,5,3,10], color='m', marker='o', markersize=6,
         linewidth=3, linestyle='dashed')

plt.grid(axis='y', color='grey', linewidth=2)
plt.show()
```



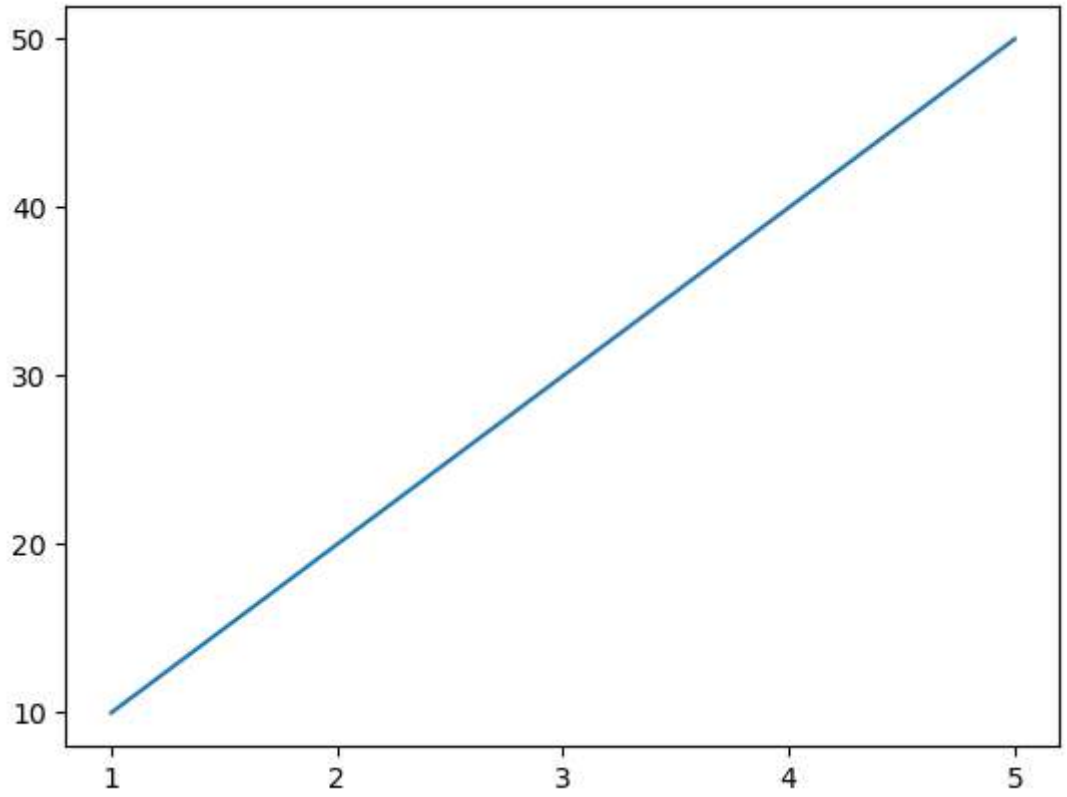
In [84]:
```

```

In [85]:
```
plt.figure(figsize=(14, 5))
plt.plot([1,2,1,5,3,1,2,1,5,3,1,2,1,5,3,1,2,1,5,3,5,4,1,2,1,5,3,5,4,5,
          1,2,1,5,3,5,4,1,2,1,5,3,5,4,5,1,2,1,5,3,10], color='m', marker='o', markersize=6,
         linewidth=3, linestyle='dashed', label='Line Plot')

plt.title('Sample Graph', fontsize=15, loc='center')
plt.xlabel('Random Number: X-Axis', fontsize=15,)
plt.ylabel('Y-Axis', fontsize=15)
plt.xticks([0, 10, 20, 30, 40, 50], ['January', 'February', 'March', 'April', 'May', 'June'])
plt.yticks(np.arange(1,11,1))
plt.grid()
plt.legend()
plt.show()
```



In [ ]:

In [90]:
```
x = [1,2,3,4,5]
y = [10, 20, 30, 40, 50]

plt.plot(x, y)
plt.xticks(np.arange(1,6,1))
plt.yticks(np.arange(10,51,10))
plt.show()
```

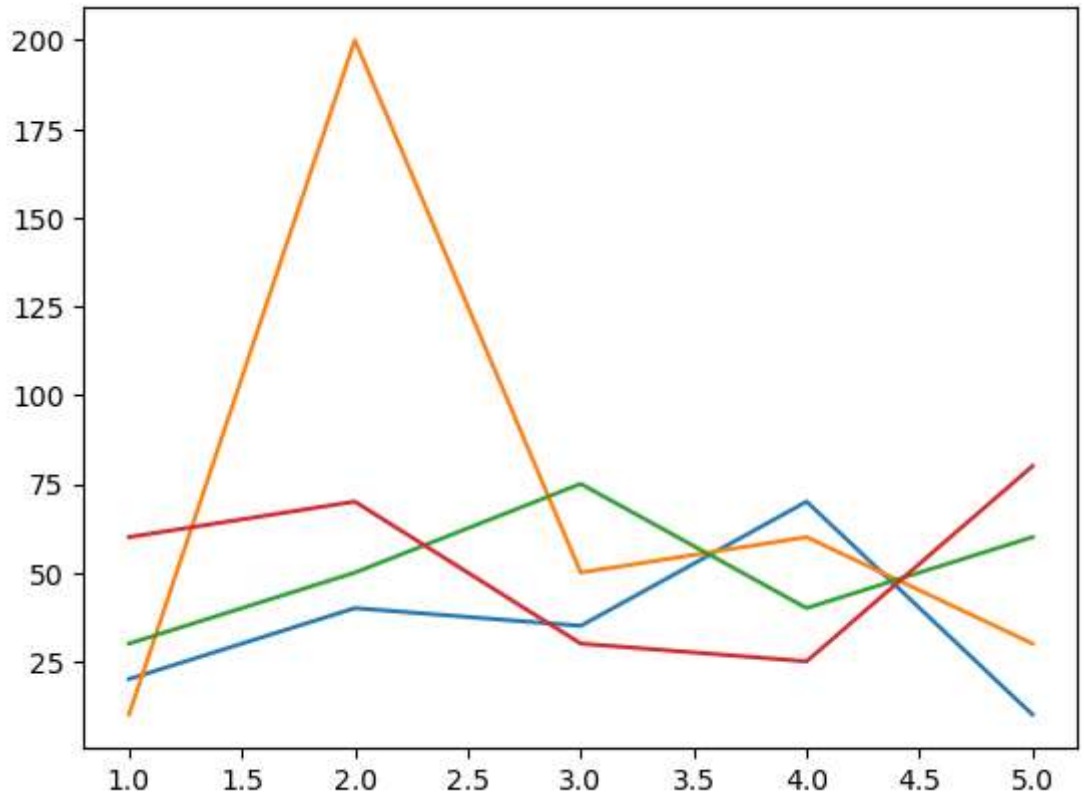


In [ ]:

In [ ]:

Line Plot

In [93]:
```
x = [1, 2, 3, 4, 5]

y1 = [20, 40, 35, 70, 10]
y2 = [10, 200, 50, 60, 30]
y3 = [30, 50, 75, 40, 60]
y4 = [60, 70, 30, 25, 80]

plt.plot(x, y1)
plt.plot(x, y2)
plt.plot(x, y3)
plt.plot(x, y4)
plt.show()
```

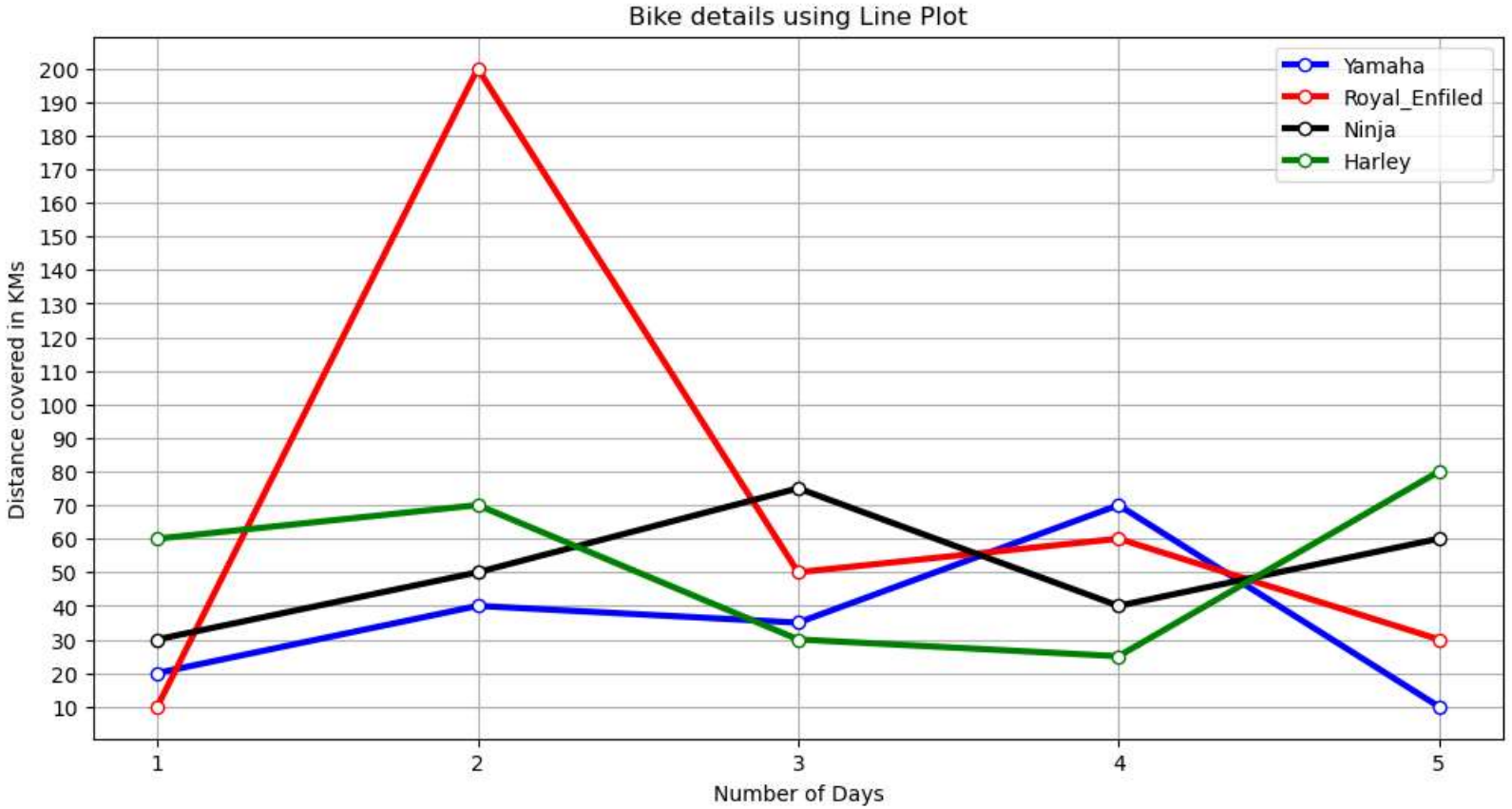


In [ ]:

```
#   marker: marker style string, `~.path.Path` or `~.markers.MarkerStyle`
#     markeredgecolor or mec: color
#     markeredgewidth or mew: float
#     markerfacecolor or mfc: color
#     markerfacecoloralt or mfcalt: color
#     markersize or ms: float
```

In [113]:

```
x = [1, 2, 3, 4, 5]

y1 = [20, 40, 35, 70, 10]
y2 = [10, 200, 50, 60, 30]
y3 = [30, 50, 75, 40, 60]
y4 = [60, 70, 30, 25, 80]

plt.figure(figsize=(12, 6))
plt.plot(x, y1, color='b', marker='o', label='Yamaha', lw=3, markerfacecolor='white')
plt.plot(x, y2, color='r', marker='o', label='Royal_Enfiled', lw=3, mfc='w')
plt.plot(x, y3, color='k', marker='o', label='Ninja', lw=3, mfc='w')
plt.plot(x, y4, color='g', marker='o', label='Harley', lw=3, mfc='w')

plt.title('Bike details using Line Plot')
plt.xlabel('Number of Days')
plt.ylabel('Distance covered in KMs')
plt.xticks(np.arange(1,6,1))
plt.yticks(np.arange(10,201,10))
plt.legend()
plt.grid()
plt.show()
```

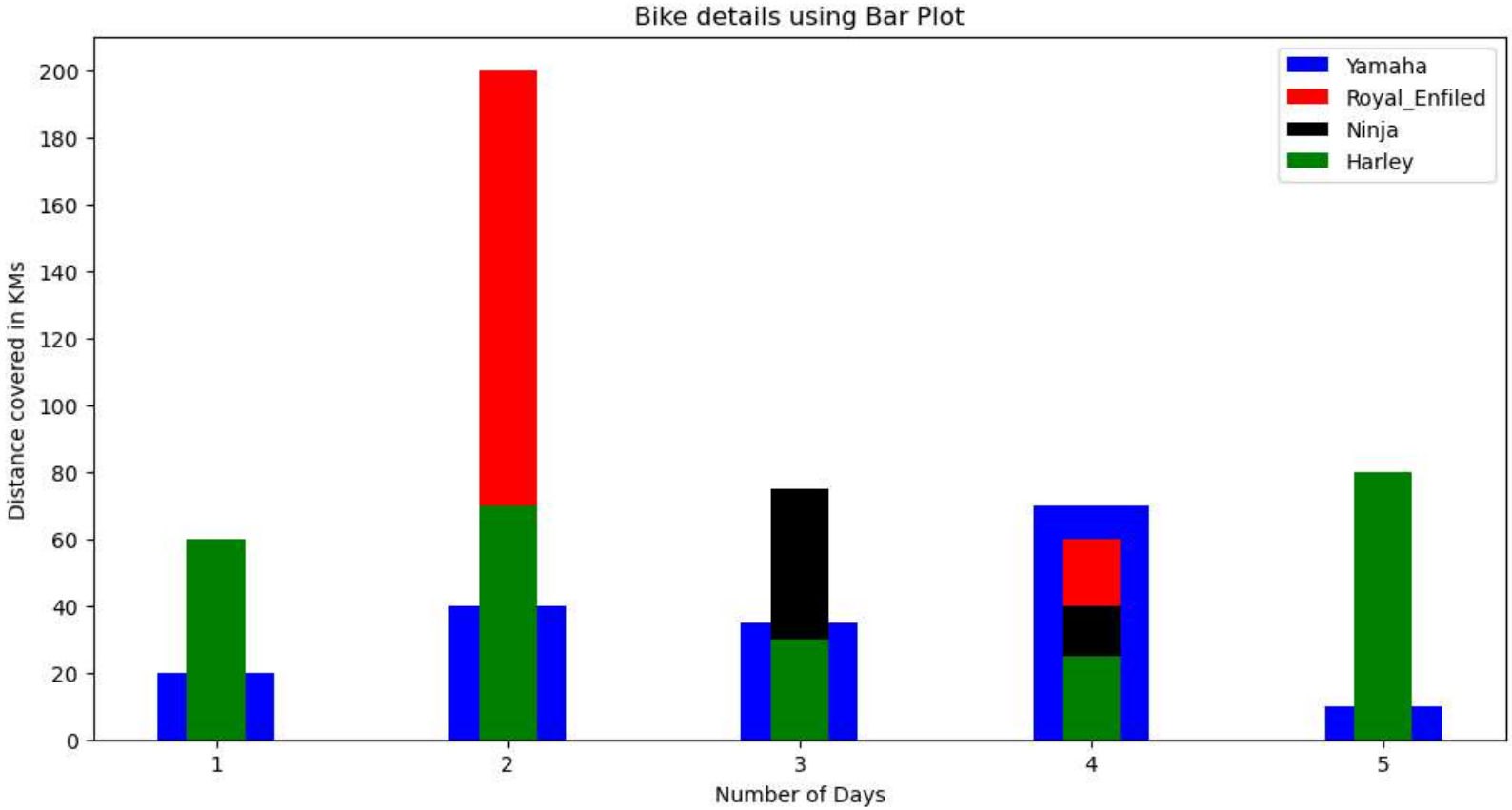

Bike details using Line Plot
Yamaha
Royal_Enfiled
Ninja
Harley
Distance covered in KMs
Number of Days
200
190
180
170
160
150
140
130
120
110
100
90
80
70
60
50
40
30
20
10
1
2
3
4
5

In [ ]:

Bar Plot

In [13]:

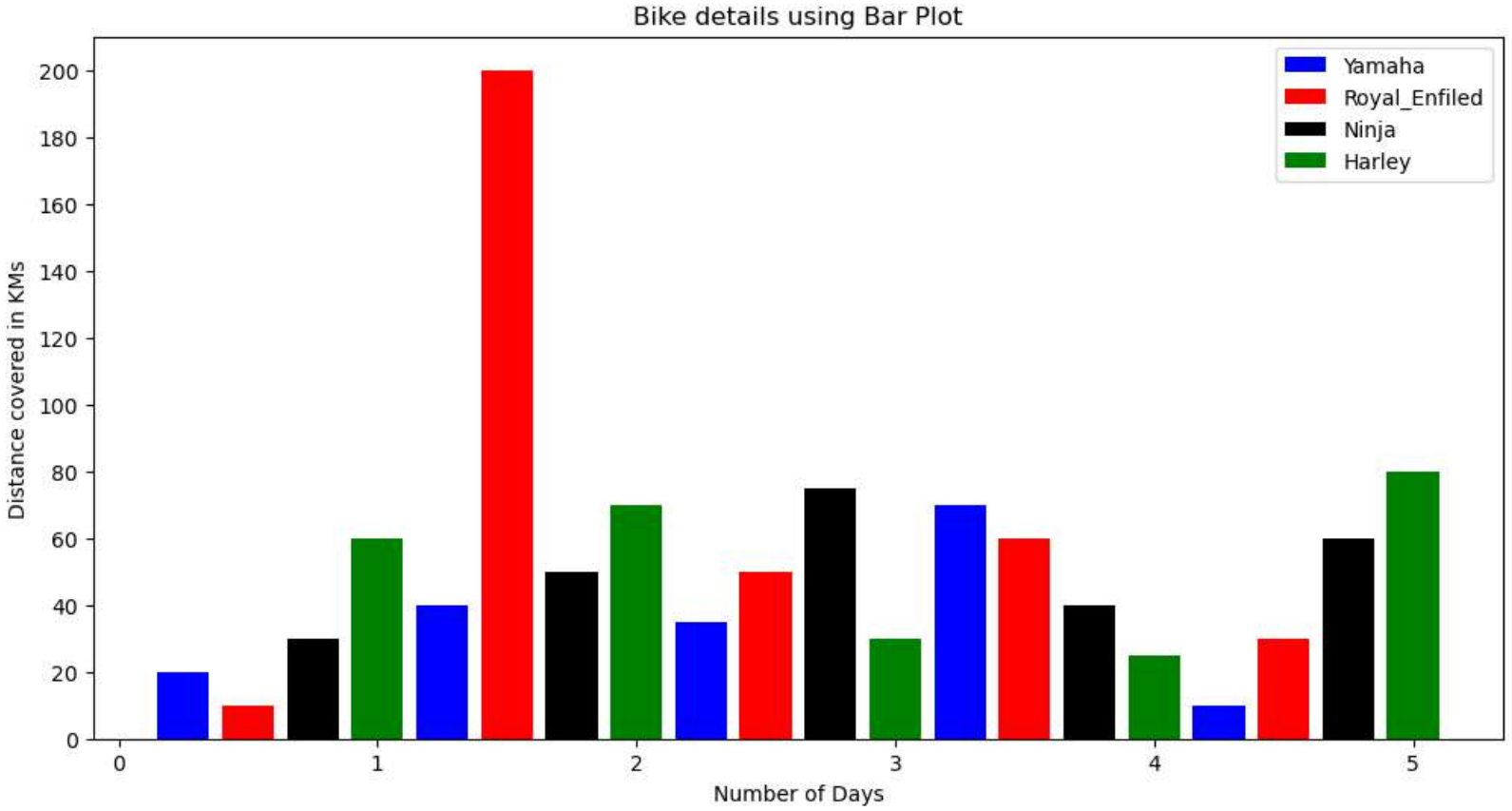
```
x = [1, 2, 3, 4, 5]

y1 = [20, 40, 35, 70, 10]
y2 = [10, 200, 50, 60, 30]
y3 = [30, 50, 75, 40, 60]
y4 = [60, 70, 30, 25, 80]

plt.figure(figsize=(12, 6))
plt.bar(x, y1, color='b', label='Yamaha', width=0.4)
plt.bar(x, y2, color='r', label='Royal_Enfiled', width=0.2)
plt.bar(x, y3, color='k', label='Ninja', width=0.2)
plt.bar(x, y4, color='g', label='Harley', width=0.2)

plt.title('Bike details using Bar Plot')
plt.xlabel('Number of Days')
plt.ylabel('Distance covered in KMs')
plt.yticks(np.arange(0,201,20))
plt.legend()
# plt.grid()
plt.show()
```


Bike details using Bar Plot
Yamaha
Royal_Enfiled
Ninja
Harley
Distance covered in KMs
0
20
40
60
80
100
120
140
160
180
200
1
2
3
4
5
Number of Days

In [ ]:

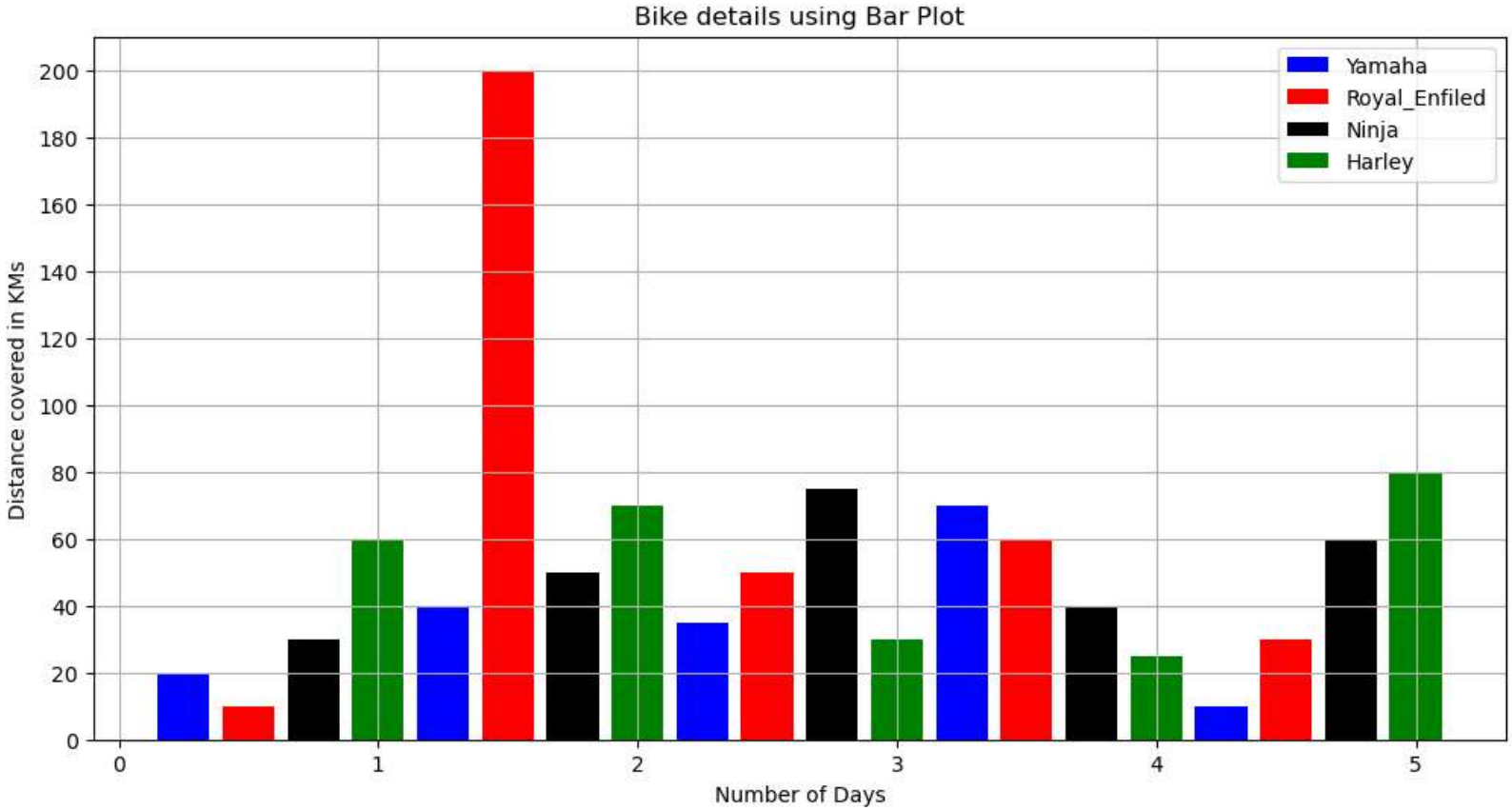
In [15]:

```
x1 = [0.25, 1.25, 2.25, 3.25, 4.25]
x2 = [0.5,  1.5,  2.5,  3.5,  4.5]
x3 = [0.75, 1.75, 2.75, 3.75, 4.75]
x4 = [1,    2,    3,    4,    5]


y1 = [20, 40, 35, 70, 10]
y2 = [10, 200, 50, 60, 30]
y3 = [30, 50, 75, 40, 60]
y4 = [60, 70, 30, 25, 80]

plt.figure(figsize=(12, 6))
plt.bar(x1, y1, color='b', label='Yamaha', width=0.2)
plt.bar(x2, y2, color='r', label='Royal_Enfiled', width=0.2)
plt.bar(x3, y3, color='k', label='Ninja', width=0.2)
plt.bar(x4, y4, color='g', label='Harley', width=0.2)

plt.title('Bike details using Bar Plot')
plt.xlabel('Number of Days')
plt.ylabel('Distance covered in KMs')
plt.yticks(np.arange(0,201,20))
plt.legend()
# plt.grid()
plt.show()
```


Bike details using Bar Plot
Yamaha
Royal_Enfiled
Ninja
Harley
Distance covered in KMs
Number of Days
0
20
40
60
80
100
120
140
160
180
200
0
1
2
3
4
5

In [16]:

```
x1 = [0.25, 1.25, 2.25, 3.25, 4.25]
x2 = [0.5,  1.5,  2.5,  3.5,  4.5]
x3 = [0.75, 1.75, 2.75, 3.75, 4.75]
x4 = [1,    2,    3,    4,    5]


y1 = [20, 40, 35, 70, 10]
y2 = [10, 200, 50, 60, 30]
y3 = [30, 50, 75, 40, 60]
y4 = [60, 70, 30, 25, 80]

plt.figure(figsize=(12, 6))
plt.bar(x1, y1, color='b', label='Yamaha', width=0.2)
plt.bar(x2, y2, color='r', label='Royal_Enfiled', width=0.2)
plt.bar(x3, y3, color='k', label='Ninja', width=0.2)
plt.bar(x4, y4, color='g', label='Harley', width=0.2)

plt.title('Bike details using Bar Plot')
plt.xlabel('Number of Days')
plt.ylabel('Distance covered in KMs')
plt.yticks(np.arange(0,201,20))
plt.legend()
plt.grid()
plt.show()
```

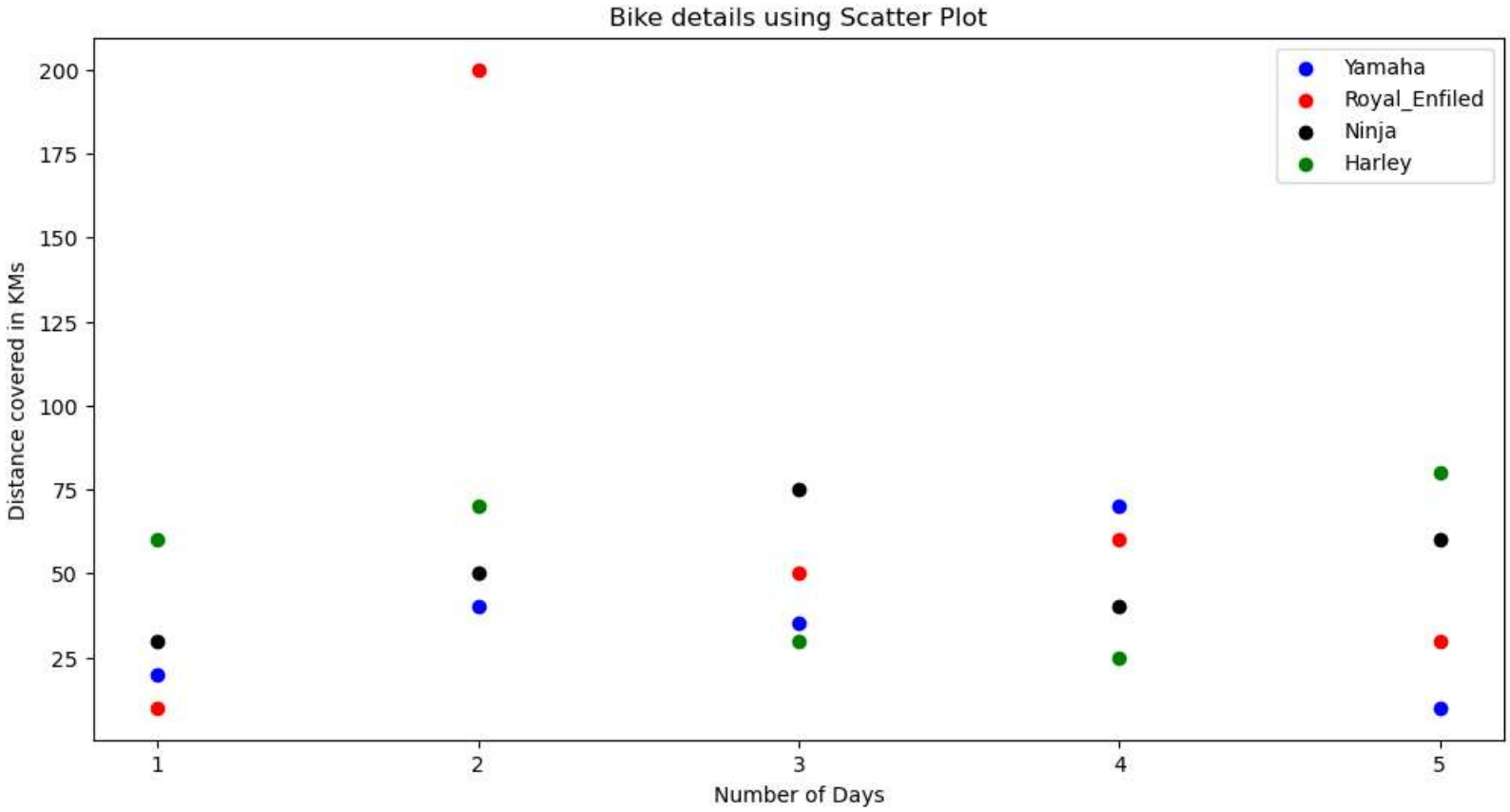


In [ ]:

Scatter Plot

In [23]:

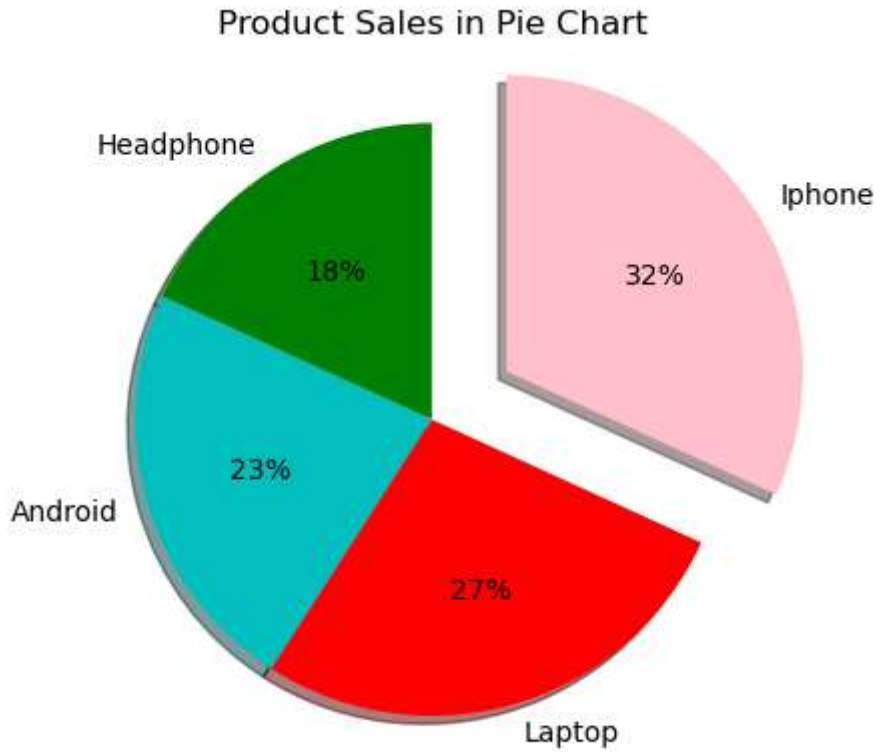
```
x = [1, 2, 3, 4, 5]

y1 = [20, 40, 35, 70, 10]
y2 = [10, 200, 50, 60, 30]
y3 = [30, 50, 75, 40, 60]
y4 = [60, 70, 30, 25, 80]

plt.figure(figsize=(12, 6))
plt.scatter(x, y1, color='b', marker='o', label='Yamaha')
plt.scatter(x, y2, color='r', marker='o', label='Royal_Enfiled')
plt.scatter(x, y3, color='k', marker='o', label='Ninja')
plt.scatter(x, y4, color='g', marker='o', label='Harley')

plt.title('Bike details using Scatter Plot')
plt.xlabel('Number of Days')
plt.ylabel('Distance covered in KMs')
plt.xticks(np.arange(1,6,1))
# plt.yticks(np.arange(10,201,10))
plt.legend()
# plt.grid()
plt.show()
```


Bike details using Scatter Plot
Yamaha
Royal_Enfiled
Ninja
Harley
Distance covered in KMs
Number of Days
200
175
150
125
100
75
50
25
1
2
3
4
5

In [ ]:

Pie Plot

In [49]:
```
slices = [4,5,6,7]
labels = ['Headphone', 'Android', 'Laptop', 'Iphone']
cols = ['g', 'c', 'r', 'pink']

plt.pie(slices, labels=labels, colors=cols, autopct='%1.0f%%', startangle=90, shadow=True, explode=(0,0,0
plt.title('Product Sales in Pie Chart')
plt.show()
```



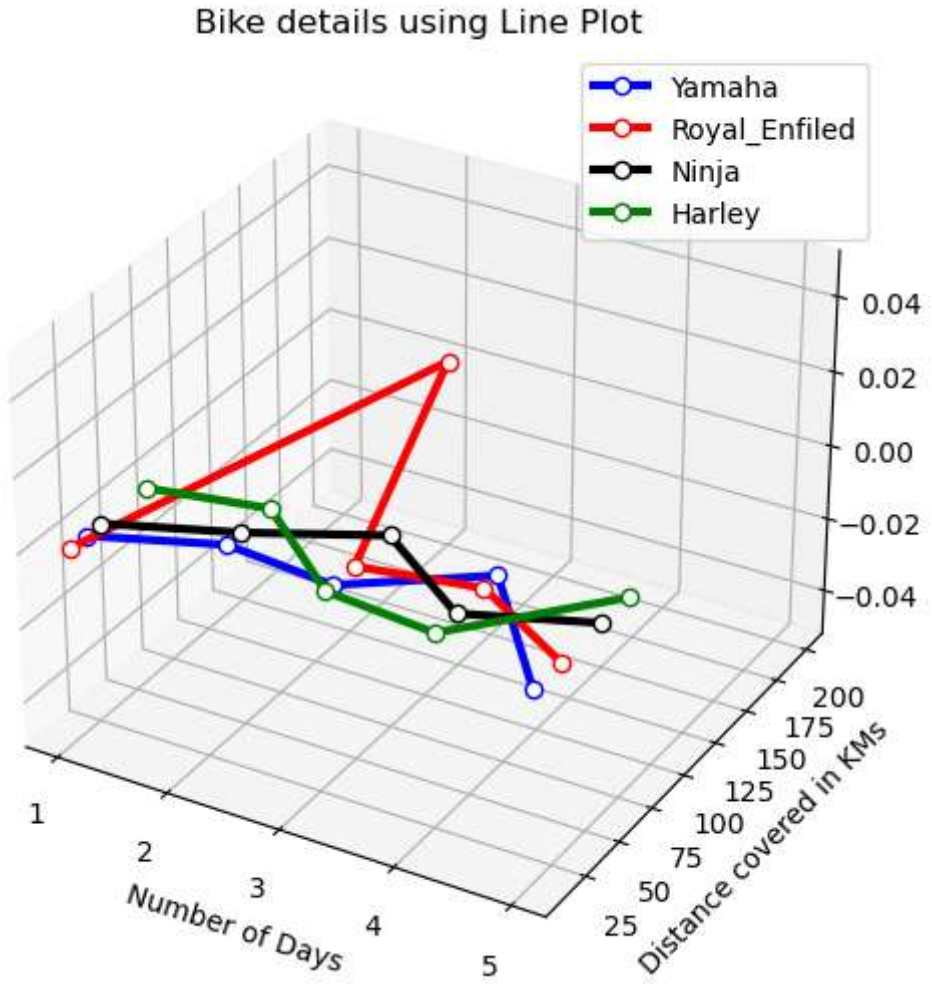
In [ ]:
```

```

3D Graph

In [ ]:

In [54]:

```
fig = plt.figure(figsize=(12, 6))
ax = fig.add_subplot(projection='3d')

x = [1, 2, 3, 4, 5]

y1 = [20, 40, 35, 70, 10]
y2 = [10, 200, 50, 60, 30]
y3 = [30, 50, 75, 40, 60]
y4 = [60, 70, 30, 25, 80]


plt.plot(x, y1, color='b', marker='o', label='Yamaha', lw=3, markerfacecolor='white')
plt.plot(x, y2, color='r', marker='o', label='Royal_Enfiled', lw=3, mfc='w')
plt.plot(x, y3, color='k', marker='o', label='Ninja', lw=3, mfc='w')
plt.plot(x, y4, color='g', marker='o', label='Harley', lw=3, mfc='w')

plt.title('Bike details using Line Plot')
plt.xlabel('Number of Days')
plt.ylabel('Distance covered in KMs')
plt.xticks(np.arange(1,6,1))
# plt.yticks(np.arange(10,201,10))
plt.legend()
# plt.grid()
plt.show()
```


Bike details using Line Plot
Yamaha
Royal_Enfiled
Ninja
Harley
0.04
0.02
0.00
−0.02
−0.04
1
2
3
4
5
Number of Days
25
50
75
100
125
150
175
200
Distance covered in KMs

In [ ]:

In [ ]:

In [ ]:

In [ ]:

Seaborn Library

In [61]:
```
import seaborn as sns
import pandas as pd
```

In [60]:
```
sns.get_dataset_names()
```

```
['anscombe',
 'attention',
 'brain_networks',
 'car_crashes',
 'diamonds',
 'dots',
 'dowjones',
 'exercise',
 'flights',
 'fmri',
 'geyser',
 'glue',
 'healthexp',
 'iris',
 'mpg',
 'penguins',
 'planets',
 'seaice',
 'taxis',
 'tips',
```

In [ ]:

In [63]:
```
df = sns.load_dataset('iris')
df
```

Out[63]:

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	virginica
146	6.3	2.5	5.0	1.9	virginica
147	6.5	3.0	5.2	2.0	virginica
148	6.2	3.4	5.4	2.3	virginica
149	5.9	3.0	5.1	1.8	virginica

150 rows × 5 columns

In [65]:
```
df['species'].value_counts()
```

Out[65]:
```
setosa        50
versicolor    50
virginica     50
Name: species, dtype: int64
```

In [68]:
```
import warnings
warnings.filterwarnings('ignore')
```

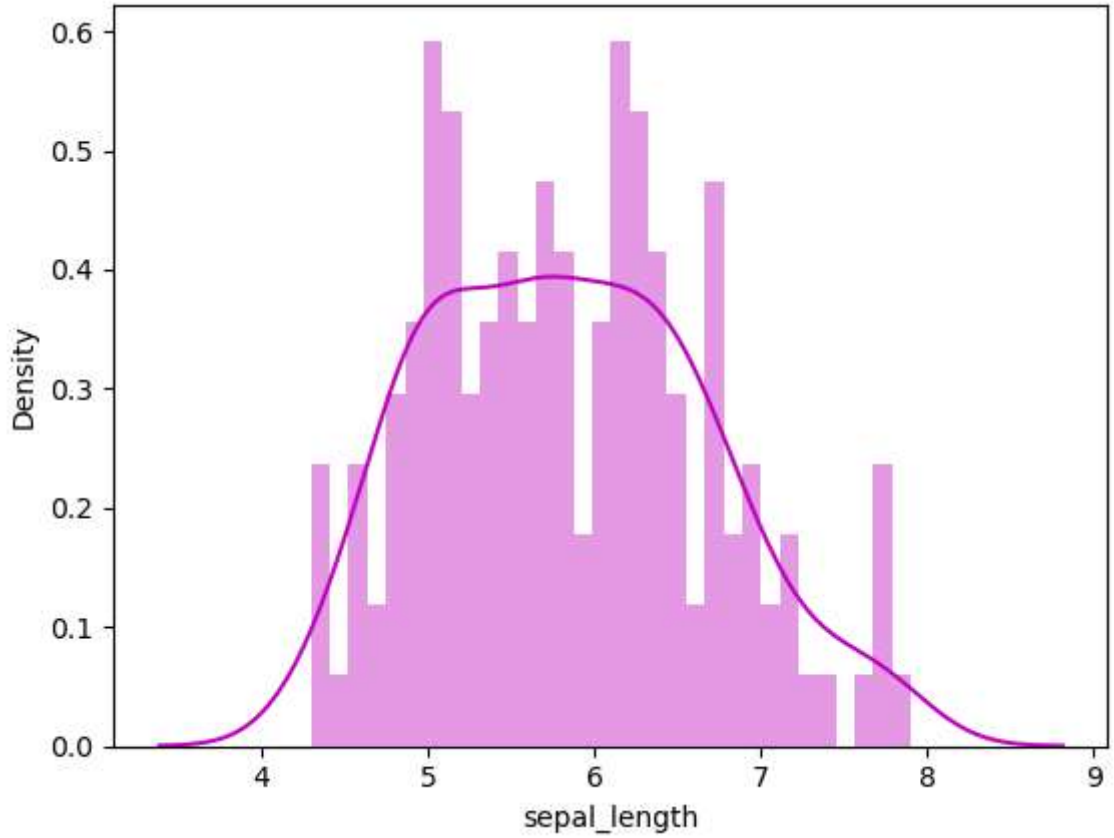
In [69]:
```
df['sepal_length']
```

Out[69]:
```
0      5.1
1      4.9
2      4.7
3      4.6
4      5.0
      ...
145    6.7
146    6.3
147    6.5
148    6.2
149    5.9
Name: sepal_length, Length: 150, dtype: float64
```

Distplot

In [82]:
```
sns.distplot(df['sepal_length'], bins=32, kde=True, color='m')
```

Out[82]: <Axes: xlabel='sepal_length', ylabel='Density'>

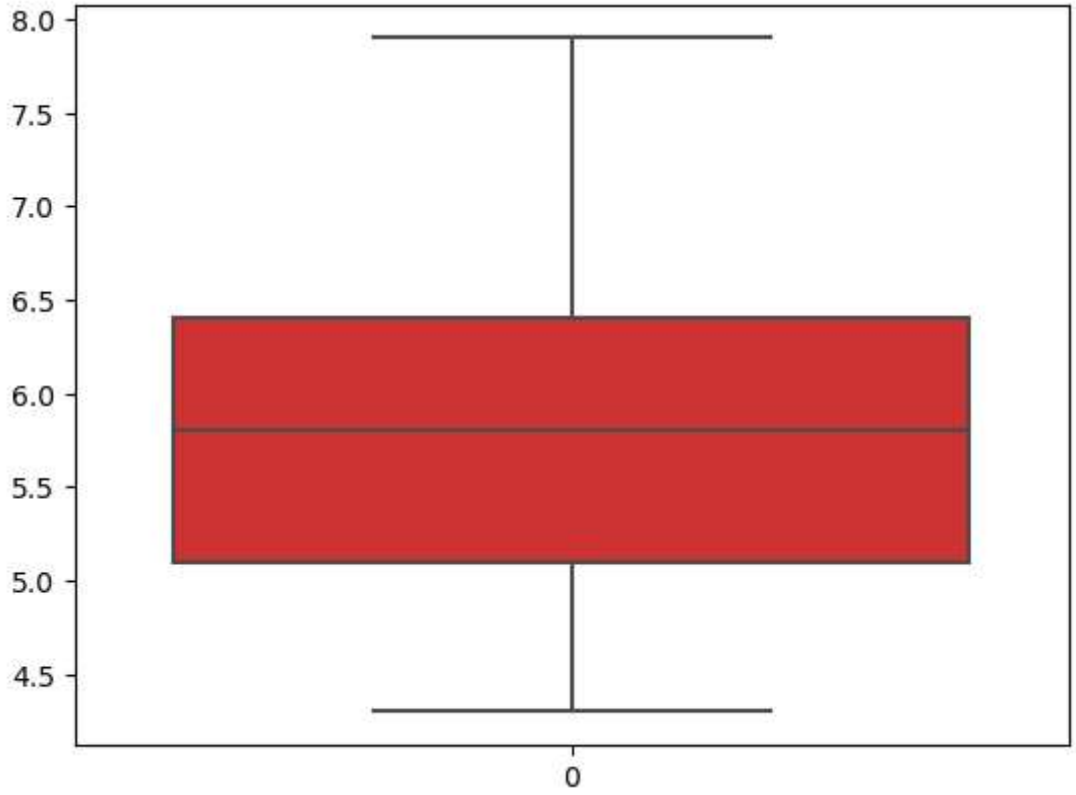


In [ ]:
```

```

Boxplot

In [78]:
```
sns.boxplot(df['sepal_length'], color='r', saturation=0.60)
```

Out[78]: <Axes: >



In [ ]:
```

```

In [72]:
```
df.describe()
```

Out[72]:

	sepal_length	sepal_width	petal_length	petal_width
count	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.057333	3.758000	1.199333
std	0.828066	0.435866	1.765298	0.762238
min	4.300000	2.000000	1.000000	0.100000
25%	5.100000	2.800000	1.600000	0.300000
50%	5.800000	3.000000	4.350000	1.300000
75%	6.400000	3.300000	5.100000	1.800000
max	7.900000	4.400000	6.900000	2.500000

In [ ]:
```

```

In [83]:
```
df.corr()
```

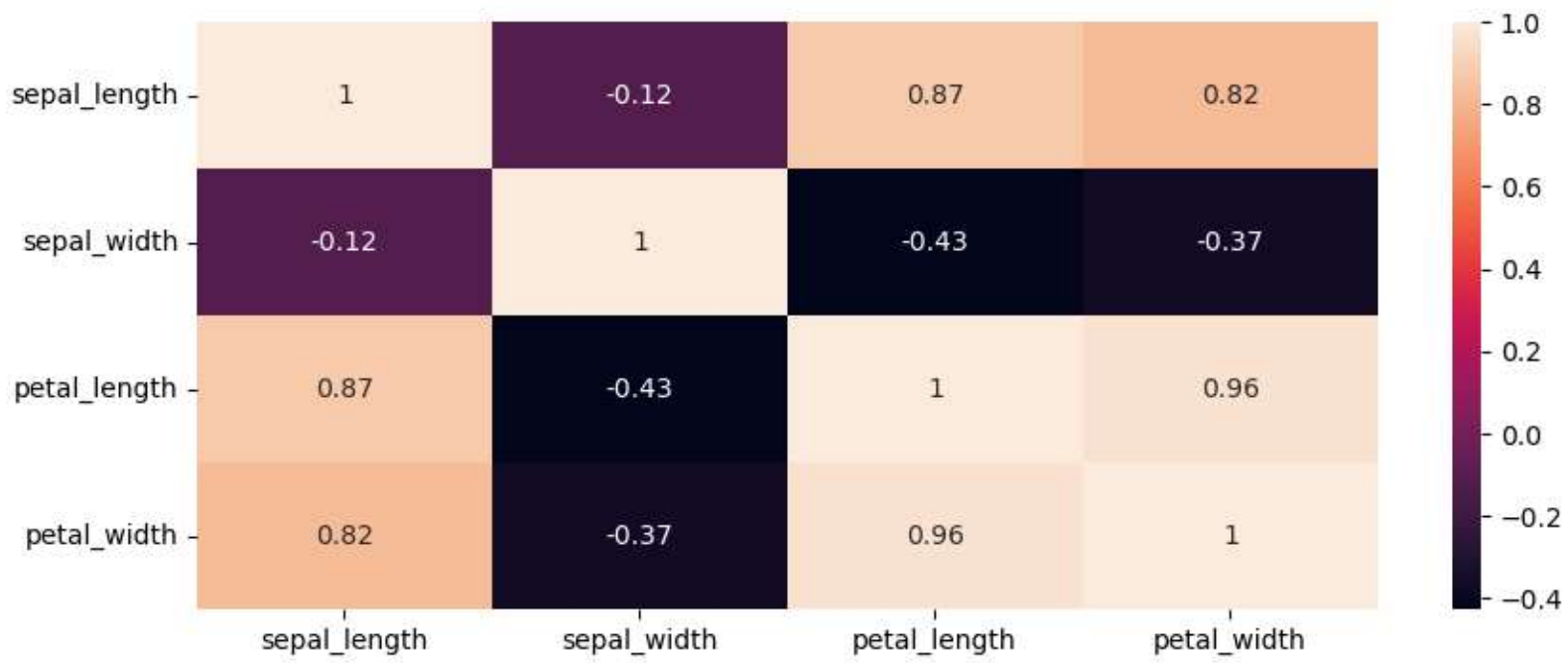
Out[83]:

	sepal_length	sepal_width	petal_length	petal_width
sepal_length	1.000000	-0.117570	0.871754	0.817941
sepal_width	-0.117570	1.000000	-0.428440	-0.366126
petal_length	0.871754	-0.428440	1.000000	0.962865
petal_width	0.817941	-0.366126	0.962865	1.000000

Heatmap

In [86]:
```
plt.figure(figsize=(10,4))
sns.heatmap(df.corr(), annot=True)
```

Out[86]: <Axes: >

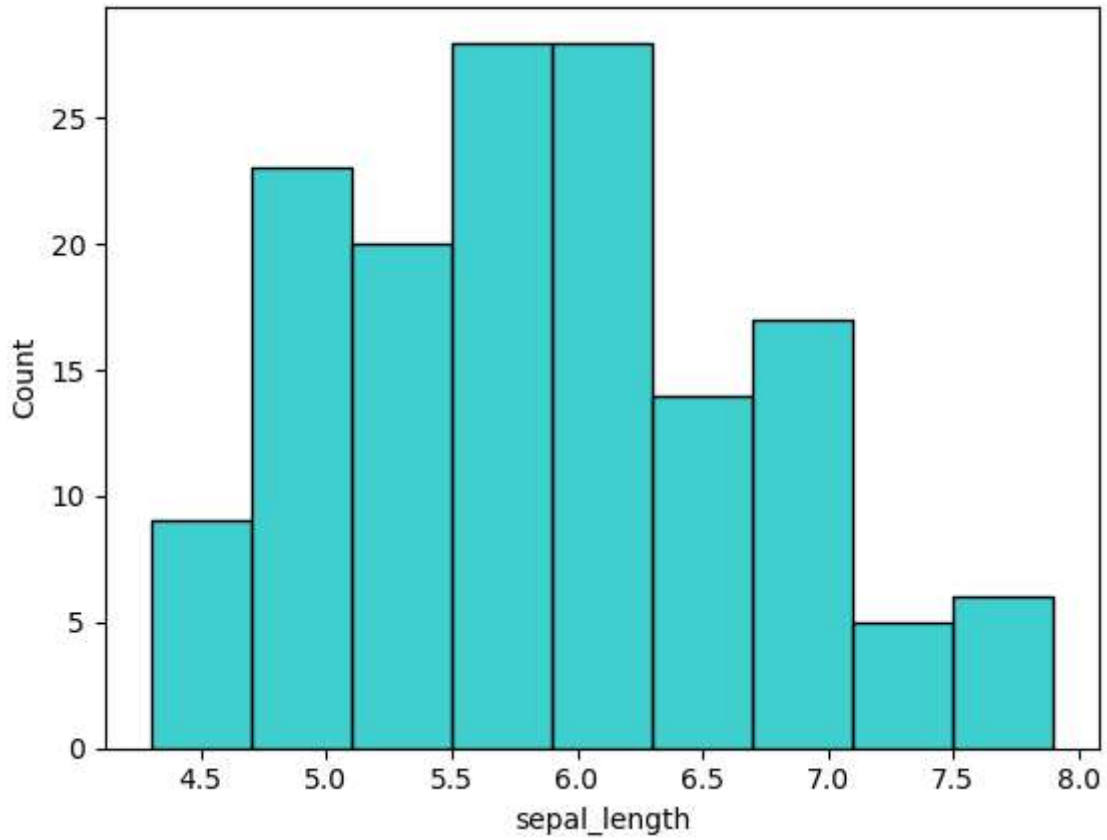


In [ ]:

Histogram

In [90]:
```
sns.histplot(df['sepal_length'], color='c')
```

Out[90]: <Axes: xlabel='sepal_length', ylabel='Count'>

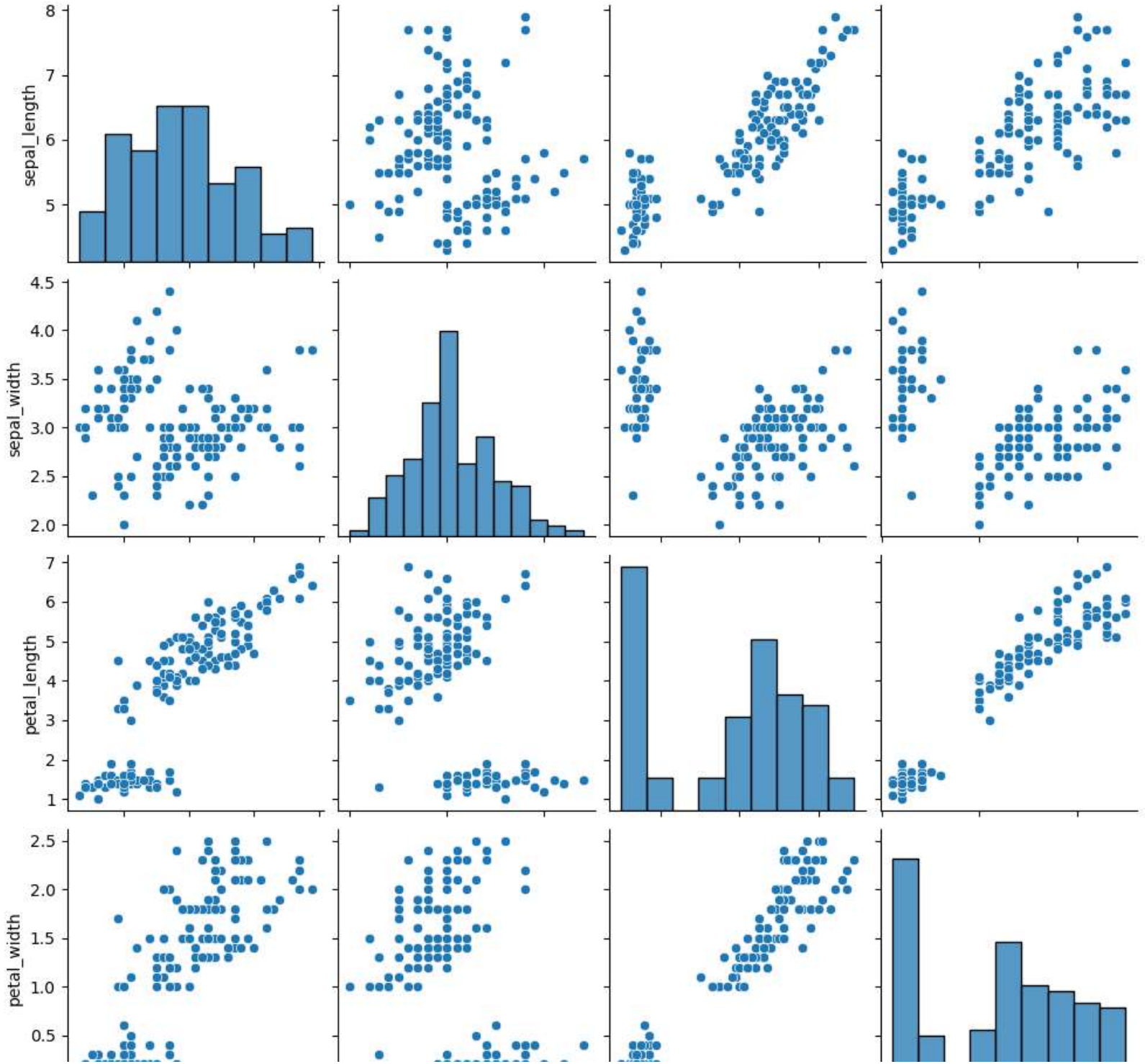


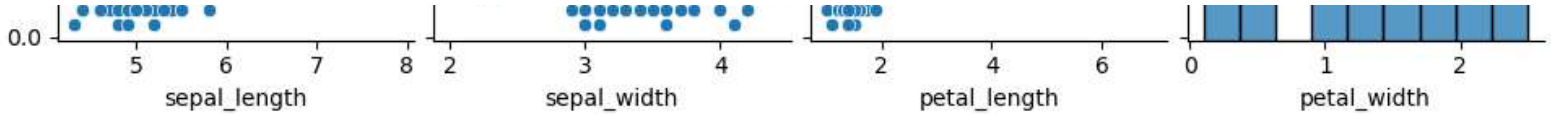
In [ ]:
```

```

Univariate Analysis

In [93]:
```
sns.pairplot(df)
```

Out[93]:
```
<seaborn.axisgrid.PairGrid at 0x21bcdd40110>
```


sepal_length
sepal_width
petal_length
petal_width



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